

Pre-screening

Participant	002	
	Yes	No
Do you or have you had a history of migraine headaches?	☐	☒
Do you or have you had a history of claustrophobia?	☐	☒
Do you or have you had a history of frequent or severe motion sickness?	☐	☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐	☒
If you are a female, are you or is there a possibility that you might be pregnant?	☐	☒

Demographics

1. What is your race/ethnicity?	☒ White / Caucasian
2. What is your gender?	☒ Male
3. How old are you?	40
4. What is the highest level of education you have completed?	☒ Some college/Associates/Technical school training
5. Which option best describes your current home?	☒ It is rented for money by you or someone in household.
6. How do you pay for healthcare?	☒ Private insurance
8. Which best corresponds to your current work situation?	☒ Working full time.
9. What is your combined family income for the past 12 months (from all sources)	☒ \$20,000- \$49,999

Timeline Followback

Visit 1

Visit 1 Date

12-11-2024

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 2

Visit 2 Date

01-02-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 3

Visit 3 Date

06-06-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 4

Visit 4 Date

07-22-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 5

Visit 5 Date

07-29-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Phone

Participant # _____

056

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☐	☒	☐	☐
2. I found the system unnecessarily complex	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☐	☒	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☒	☐	☐	☐	☐
2. The time I spent in this app just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the app task.	☒	☐	☐	☐	☐
4. I felt frustrated while using the app.	☒	☐	☐	☐	☐

5. I found this app to be confusing.	☒	☐	☐	☐	☐
6. Using this app was taxing.	☒	☐	☐	☐	☐
7. The app is attractive.	☐	☐	☐	☒	☐
8. The app is aesthetically appealing.	☐	☒	☐	☐	☐
9. The app appealed to my senses.	☐	☐	☒	☐	☐
10. Using this app was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were videos easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

AR

Participant ID _____

002

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ It is rented for money by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGROUP Presence Questionnaire

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Agree |
| 2. I felt that the virtual world surround me. | ☒ Somewhat Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Neither Agree nor Disagree |
| 6. I felt present in the virtual space | ☒ Completely Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Somewhat Disagree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Agree |
| 11 How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Neither Agree nor Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☒	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☒	☐	☐	☐
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☒	☐	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☒	☐	☐	☐	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☒	☐	☐

8. The augmented reality is aesthetically appealing.	☒	☐	☐	☐	☐
9. The augmented environment appealed to my senses.	☐	☐	☒	☐	☐
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☒	☐	☐
12. I felt interested in the therapy tasks.	☐	☐	☒	☐	☐

iPAL Application and Intervention Evaluation AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☒	☐	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒

VR

Participant ID _____ 002

1. Which option describes your current home? ☒ It is rented for money by you or someone in household

2. How do you pay for healthcare? ☒ Private insurance

3. Which best corresponds to your current work situation? ☒ Working full time.

iGroup Presence Scale

1. In the computer-generated world, I had a sense of being there. ☒ Somewhat Agree

2. I felt that the virtual world surrounded me. ☒ Completely Agree

3. I felt like I was perceiving pictures. ☒ Somewhat Disagree

4. I did not feel present in the virtual space. ☒ Completely Disagree

5. I had a sense of acting in the virtual space, rather than operating something from the outside. ☒ Completely Agree

6. I felt present in the virtual space. ☒ Completely Agree

7. How aware were you of the real world surrounding you while navigating in the virtual world? ☒ Somewhat aware

8. I was not aware of my real environment. ☒ Somewhat Disagree

9. I still paid attention to my real environment. ☒ Completely Disagree

10. I was completely captivated by the virtual world. ☒ Completely Agree

11. How real did the virtual world seem to you? ☒ Neither real nor unreal

12. How much did your experience in the virtual environment seem consistent with your real-world experience? ☒ Somewhat Agree

13. How real did the virtual world seem to you? ☒ Neither real nor unreal

14. The virtual world seemed more realistic than the real world. ☒ Completely Disagree

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this virtual experience	☐	☐	☒	☐	☐
2. The time I spent in the virtual world just slipped away.	☐	☐	☒	☐	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The virtual environment is aesthetically appealing.	☐	☐	☐	☐	☒
9. The virtual environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Evaluation VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☒	☐	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

MR

Participant ID _____

002

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ It is rented for money by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGROUP Presence Questionnaire MR

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Agree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat disagree |
| 4. I did not feel present in the virtual space. | ☒ Completely disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Neither Agree nor Disagree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11. How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Agree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely disagree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☒	☐	☐	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☒	☐	☐	☐	☐
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☒	☐	☐	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☒	☐	☐	☐	☐
8. I found the system very cumbersome to use.	☐	☒	☐	☐	☐
9. I felt very confident using the system.	☐	☒	☐	☐	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☐	☒	☐	☐	☐
2. The time I spent in mixed reality just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☐	☐	☐	☒	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☒	☐	☐

8. The mixed-reality is aesthetically appealing.	☐	☒	☐	☐	☐
9. The blended environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☒	☐	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒

Exit Interview

1. Which option describes your current home?	☒ It is rented for money by you or someone in household
2. How do you pay for healthcare?	☒ Self-pay, out of pocket
3. Which best corresponds to your current work situation?	☒ Working full time.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	VR - liked the most; immersed you in a task and made you feel like you were a part of it and not just looking at the setting. Did not enjoy AR, felt like it was disorienting and it was difficult to focus on the task with the outside world. Did not enjoy the technology of MR and does not feel like it is caught up. Did not enjoy the functionality.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	Yes, only for VR. States that as an individual who is a recovering addict, when you feel like you are a part of something like he did when he was experiencing VR, then it makes recovery easier and makes you feel like you are not the problem. Also felt the most security in VR and didn't feel judged. Felt like someone was actually trying to help him.
3. How would you describe the iPAL application, in terms of a cravings management tool?	Feels like it's up to the patient to identify if they want help and the app would primarily be beneficial to those who are wanting help. It's good to have the app as a cravings management tool because it is also convenient, as a lot of people are on their phones these days.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	CBT - felt like the videos were undermining and made it seem like the producers think that "addicts are stupid." The way the woman in the videos spoke did not seem genuine. HRV - participant notes that he personally never really got anything out of breathing exercises ever, so he's not the best to talk about his experience with this particular intervention
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? If NO, why not?	yes - feels like it would be beneficial to have the app available to individuals after a 30-day rehab program (for example) and have the app available to continue treatment and progressing towards sobriety outside of the facility setting.
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	yes - would use on a daily basis. Believes that one would have to use it daily if they are wanting to remain clean.